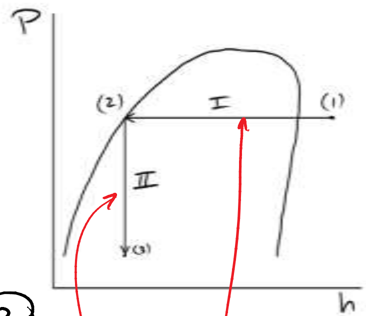
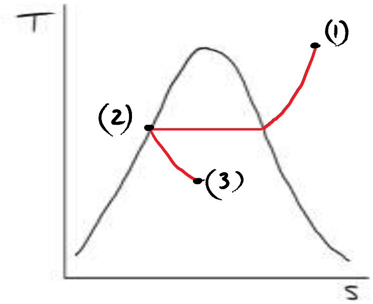
ME-ID Solution

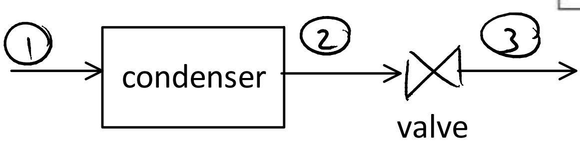
"By initialing below I agree to the following: I will not discuss this exam with others who may be taking it at a different time due to schedule conflicts, and that I will not copy off of another student's exam, both of which are considered cheating. I understand that if I am caught cheating on this exam I will receive a score of zero." Initials:

Justify your answers mathematically or in words. Answers without justification will not receive credit.

1. (4 Points) Steam flows through Device I and then through Device II. Draw the two devices based on the state transitions in the P-h diagram. Also draw the state transitions on the T-s diagram.





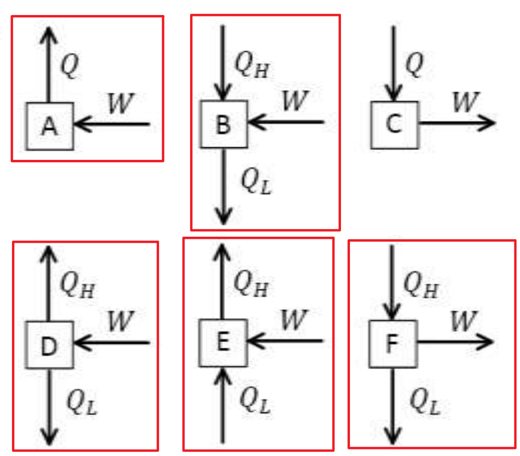


isobaric heat removal (condenser)

isenthalpic pressure reduction (valve)

2. (3 Points) Circle the processes that can run under continuous, steady-state operation.

It is possible to convert work to heat (like friction), making A, B and D possible.
Case C is impossible because heat cannot be converted fully into work.
E is a refrigerator and F is a heat engine.



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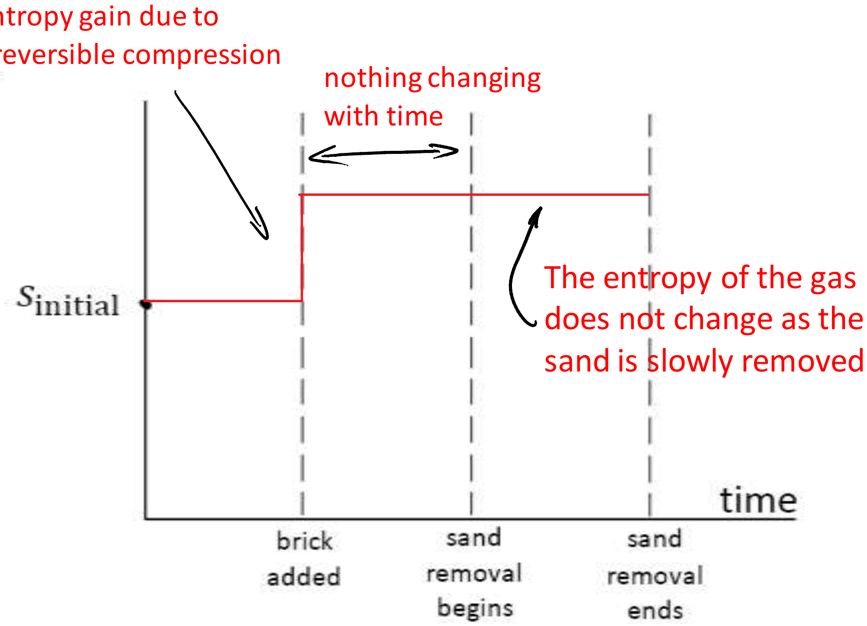
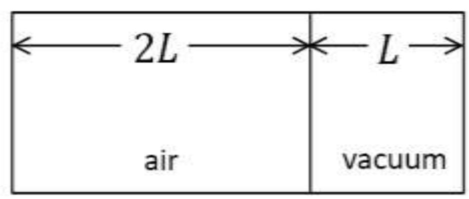
3. (3 Points) An adiabatic piston-cylinder device contains an ideal gas. The piston initially has sand piled on top of it. A brick is then added to the sand. The sand is slowly removed, leaving only the brick on top. Draw a plot of the entropy of the gas as a function of time.



4. (2 Points) Air is contained in part of a tank. The partition in the tank is removed. What happens to the enthalpy of the air before and after this process?

a. $h_2 = \frac{1}{3}h_1$
b. $h_2 = \frac{2}{3}h_1$
c. $\boxed{h_2 = h_1}$
d. $h_2 = 2h_1$
e. $h_2 = 3h_1$



$\Delta U = \cancel{Q}^{\,0} - \cancel{W_b}^{\,0}$ expands against vacuum

$\Delta U = 0 = m C_v \Delta T$

$\Delta T = 0$

$\Delta H = m C_p \Delta T = 0$

5. (2 Points) A flexible balloon filled with air is heated at ground level. It is then raised high into the atmosphere until its temperature falls back to its original temperature before it was heated. The balloon is adiabatic. The final entropy of the gas is [**greater than** / the same as / less than] the initial entropy of the gas.

$s_2 - s_1 = \cancel{C_p \ln \frac{T_2}{T_1}}^{\,0} - R \ln \frac{P_2}{P_1}$

$P_2 < P_1 \quad \ln \frac{P_2}{P_1} < 0 \rightarrow s_2 > s_1$

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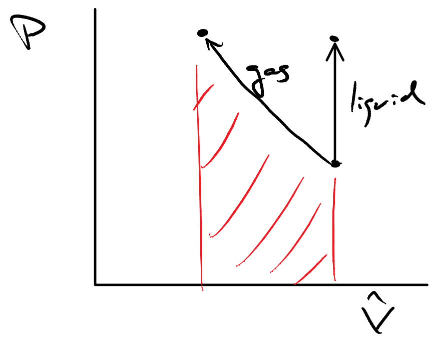
6. (2 Points) Does it require more work to adiabatically compress a saturated liquid or an ideal gas from 10 kPa to 1000 kPa? If so, which one requires more work?

a. Yes, the saturated liquid.

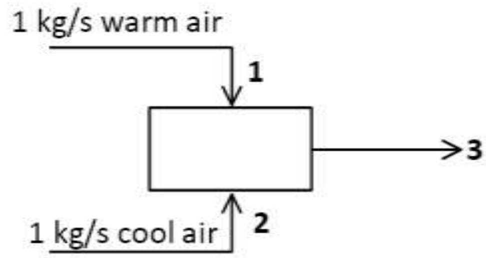
b. $\boxed{\text{Yes, the ideal gas.}}$

c. No, the same amount of work is needed in both cases.



liquid is essentially incompressible. No area under the P-V curve, no work was done on it. However, boundary work was done on the gas (shaded in red).

7. (2 Points) The streams of air are mixed isobarically.
Circle the correct sign: $s_1 + s_2 \; > \; = \; \boxed{<} \; 2\,s_3$

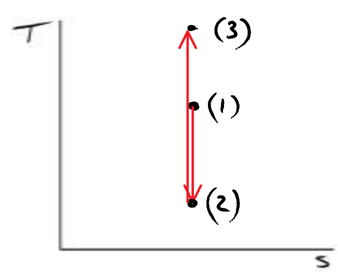


$$\cancel{\dot m}\, s_1 + \cancel{\dot m}\, s_2 + \frac{\dot S_{gen}}{\dot m} = 2\,\cancel{\dot m}\, s_3$$

$$s_1 + s_2 + \frac{\dot S_{gen}}{\dot m} = 2\, s_3$$

Mixing is an irreversible process. Entropy was generated, $\dot S_{gen} > 0$.

8. (2 Points) Starting from the top of a mountain (State 1), an adiabatic piston-cylinder device filled with air is carried high up into the atmosphere (State 2). It is then lowered to sea level (State 3). Draw the state transitions on the T-s diagram.



The gradual expansion and contraction of the gas within the cylinder is reversible and adiabatic. Therefore, the gas encounters no entropy change. Temperature decreases as it rises because the device is expanding, doing work on the atmosphere. Work is done on the device by the atmosphere as it is returned to the ground, increasing its temperature.

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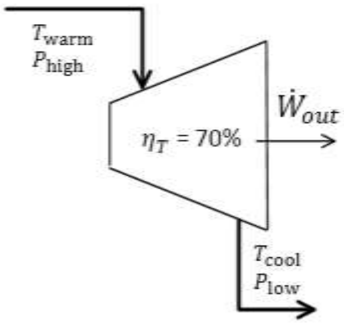
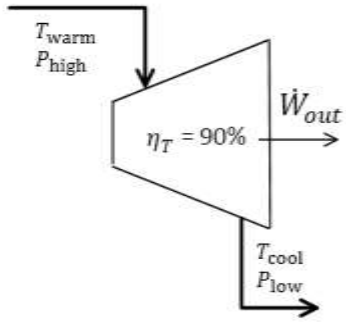
9. (3 Points) An ideal gas flows through the two turbines. Both turbines generate the same amount of shaft work. They both have the same inlet temperature, T_{warm}, and the same exhaust temperature, T_{cool}. In both turbines P_{high} and P_{low} are also the same. However, their isentropic efficiencies are different. Circle choice **a.** or **b.** and explain why.

a. This is possible.

b. $\boxed{\text{This is impossible.}}$





$$\underbrace{\dot{W}_{out}}_{\text{Same}} = \dot{m}\,\overbrace{(T_1 - T_2)}^{\text{Same}} \quad \therefore \text{ Same } \dot{m}$$

$$\dot{W}_{out,s} = \dot{m} C_p (T_1 - T_{2s}) \quad \therefore \text{ Same } \dot{W}_{out,s}$$

$$\text{Same} \rightarrow T_{2s} = T_1 \left(P_2/P_1\right)^{k-1/k}$$

$$\eta_T = \frac{\dot{W}_{out}}{\dot{W}_{out,s}}$$

$\therefore$ η_T must be same

10. (3 Points) When operated adiabatically, which device, if any, can cause the following state transition for an ideal gas? Circle all that apply:

a. throttling valve

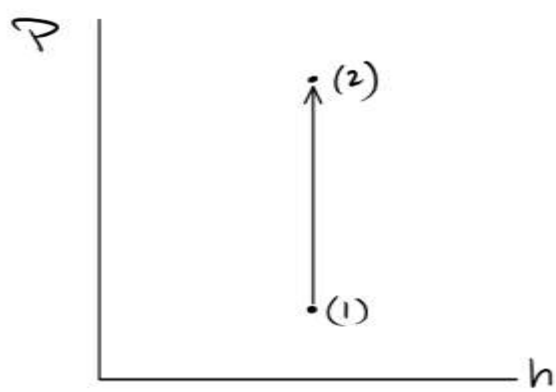
b. nozzle

c. diffuser

d. turbine

e. compressor

f. $\boxed{\text{none of the above}}$



$$\Delta h = C_p \Delta T = 0 \rightarrow T_2 = T_1$$

$$s_2 - s_1 = \underbrace{\cancel{C_p \ln \frac{T_2}{T_1}}}_{0} - R \ln \frac{P_2}{P_1} < 0$$

$$s_2 < s_1$$

impossible for entropy to decrease if the process is adiabatic

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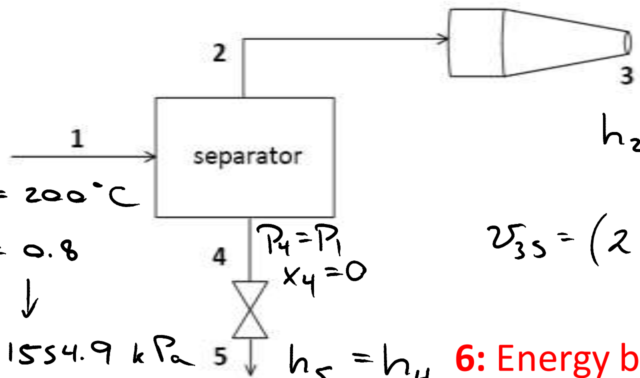
11. (25 Points) Steam at a temperature of 200 °C and a quality of 80% flows into the separator, which separates it into saturated liquid and vapor. The vapor from the separator flows through the nozzle, exiting at 100 kPa. The inlet diameter of the nozzle is considerably larger than its exit diameter. The nozzle has a relatively low isentropic efficiency. Steam exits the nozzle at a speed of 300 m/s. The liquid from the separator flows through a throttling valve and exits at 100 kPa.

a. Label the locations of all five streams on the P-h diagram. I've added one or more extra points; cross them out. I've also omitted one or more points; draw them in and label them.

b. How fast would the steam exit the nozzle if it operated reversibly? Use values directly from the diagram, if you like.

c. What is the state of the steam exiting the throttling valve? If it is in vapor-liquid equilibrium, what is its quality?



$D_2 >> D_1 \rightarrow v_2 << v_3$ **3:** Inlet v about zero

$h_2 + \frac{1}{2}\cancel{v_2^2} = h_{3s} + \frac{1}{2}v_{3s}^2$ ($v_2^2 \approx 0$) **6:** Energy balance nozzle

$v_{3s} = \left(2\left(2792 - 2330\right)\cdot 1000 \frac{m^2/s^2}{kJ/kg}\right)^{1/2} = \underline{960\ m/s}$

1: All answers correct

$T_1 = 200°C$

$x_1 = 0.8$

$\downarrow$

$P_1 = 1554.9\ kPa$

2: separator pressure

$P_4 = P_1$

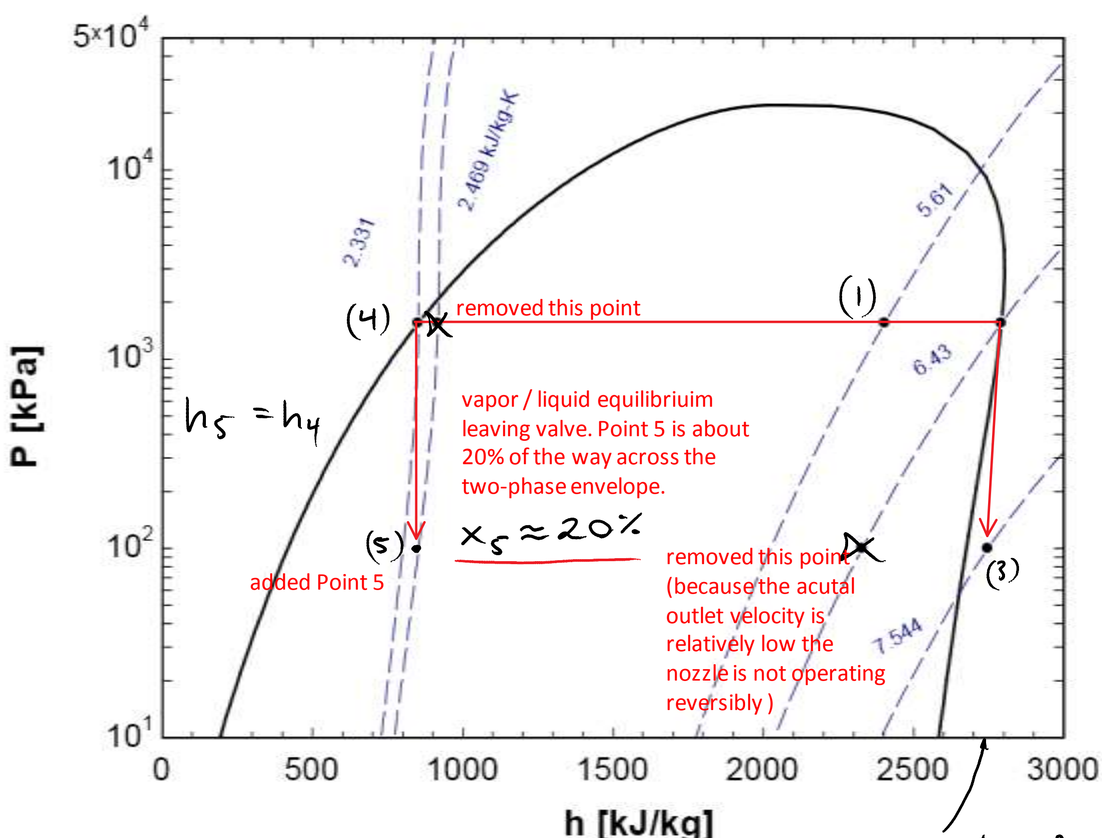
$x_4 = 0$

$h_5 = h_4$ **6:** Energy balance valve

$\left.\begin{matrix} P_2 = P_1 \\ x_2 = 1 \end{matrix}\right\} \begin{matrix} T_2 = T_1 = 200°C \\ h_2 = 2792\ kJ/kg \end{matrix}$

$\left.\begin{matrix} s_3 = s_2 = 6.43\ kJ/kg\text{-}K \\ P_3 = 100\ kPa \end{matrix}\right\} h_{3s} = 2330\ kJ/kg$

2: Enthalpy if isentropic



5: Correct P-h diagram

$h_3 = h_2 - \frac{1}{2}v_2^2 = 2747\ kJ/kg$

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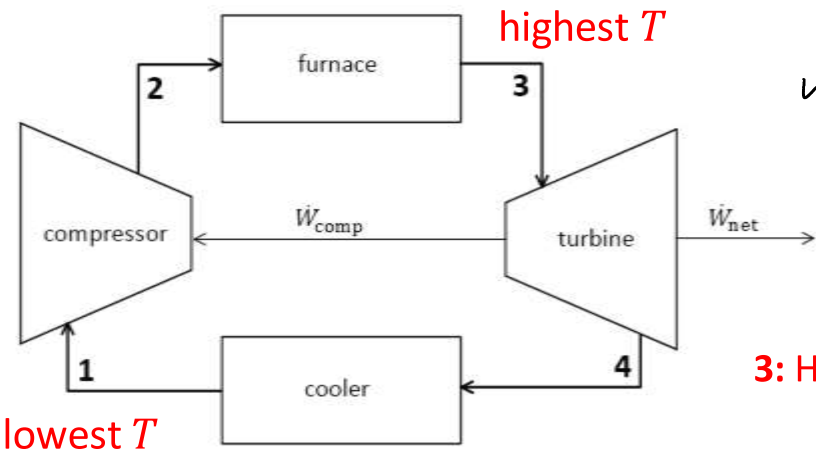
12. (25 Points) Air flows through the system at a rate of 5 kg/s. Heat flows into the system from the furnace. Heat leaves the system through the cooler. Part of the total amount of shaft work generated by the turbine is used to drive the compressor. The pressure exiting the compressor is 1 MPa. The pressure exiting the turbine is 100 kPa. The highest temperature in the system is 2000 °C. The lowest temperature in the system is 40 °C. Assume air is an ideal gas. Use the following values, and assume they are independent of temperature: c_p = 1 kJ/kg-K, R = 0.3 kJ/kg-K, and k = 1.4.

a. How much work is needed to drive the compressor ($\dot{W}_{comp}$)? It is adiabatic and reversible.

b. The turbine is adiabatic with an isentropic efficiency of η_T = 65%. How much net shaft work does it generate ($\dot{W}_{net}$)?



5: Energy balance compressor

$$\dot{W}_{comp,in} = \dot{m}(h_2 - h_1) = \dot{m} c_p (T_2 - T_1)$$

$$= 5\ \mathrm{kg/s} \cdot 1.0\ \mathrm{kJ/kg\,K}\ (604 - 313)\ \mathrm{K}$$

$$= \underline{1455\ \mathrm{kW}}$$

3: Heat capacity to relate h and T

3: Isentropic relationship compressor

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = 313\ \mathrm{K} \cdot 10^{0.4/1.4} = 604\ \mathrm{K}$$

5: Energy balance turbine

$$\dot{W}_{out,s} = \dot{m} c_p (T_3 - T_{4s}) = 5\ \mathrm{kg/s} \cdot 1.0\ \mathrm{kJ/kg\,K} \cdot (2273 - 1177)\ \mathrm{K} = 5480\ \mathrm{kW}$$

$$T_{4s} = T_3 \left(\frac{P_4}{P_3}\right)^{\frac{k-1}{k}} = 2273\ \mathrm{K} \cdot \left(\frac{1}{10}\right)^{0.4/1.4} = 1177\ \mathrm{K}$$

3: Isentropic relationship turbine

$$\dot{W}_{out} = \eta_T\, \dot{W}_{out,s} = 0.65 \cdot 5480\ \mathrm{kW} = 3562\ \mathrm{kW}$$

3: Isentropic efficiency

$$\dot{W}_{net} = \dot{W}_{out} - \dot{W}_{comp} = 3562\ \mathrm{kW} - 1165\ \mathrm{kW} = \underline{2397\ \mathrm{kW}}$$

2: Shaft work is split

1: All answers correct